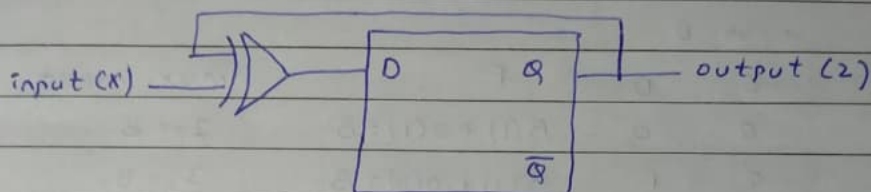


1) Explain the behavior of the circuit.



→ • Excitation Input: The input to the D-flipflop comes from the XOR gate $D = X \oplus Q$

• Next State Equation: since, as D-flipflop always copies it's input to the next state ($Q_{next} = D$) we get $Q_{next} = X \oplus Q$.

Present state (Q)	External input	next state Q_{next}	output $Z = Q$
0	0	0	0
0	1	1	0
1	0	1	1
1	1	0	1

• when $x=0$, the state remains unchanged (Hold)

• when $x=1$, the state reverses (Toggle)

2) Design a posedge triggered D-flipflop using 2:1 mux

→ To design a posedge triggered D-flipflop we use Five 2:1 muxes.

• 1-mux Configured as an inverter to generate the $\overline{clk}$ signal.

• The 4-muxes Configured in a master-slave latch.

* The 2:1 mux can be converted into a NOT gate by connecting the select line (s) to clk & setting $I_0 = 1$ (VCC) and $I_1 = 0$ (GND) as inputs.

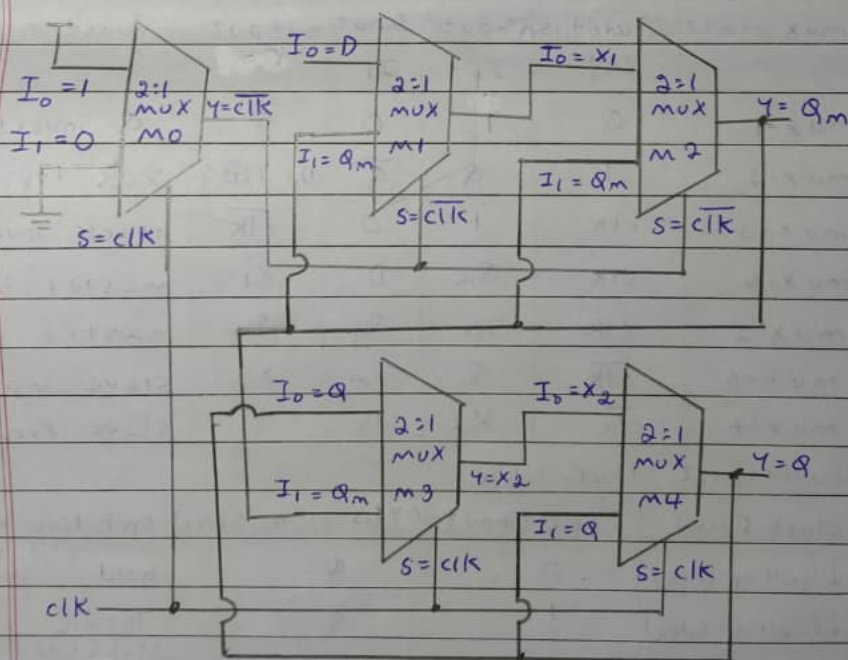
$$\text{output} = \overline{clk} \cdot (1) + clk \cdot (0) = \overline{clk}$$

mux implementation

mux state	select line (S)	Input (I_0)	Input (I_1)	output	Function
mux-0	clk	1	0	$\overline{\text{clk}}$	generate $\overline{\text{clk}}$
mux-1	$\overline{\text{clk}}$	0	Q_m	X_1	master Input Gate
mux-2	$\overline{\text{clk}}$	X_1	Q_m	Q_m	master feedback loop
mux-3	clk	Q	Q_m	X_2	slave Input Gate
mux-4	clk	X_2	Q	Q	slave feedback loop

functional truth Table

clk	data i/p (D)	next state (Q_{next})	operating mode
$\uparrow$ (Rising edge)	0	0	Reset
$\uparrow$ (Rising edge)	1	1	set
0 (low level)	X	Q	hold (master-track, slave isolate)
1 (high level)	X	Q	hold (master-isolate, slave locked)



3) Design a negedge Triggered T-flipflop using 2:1 MUXES only.

→ To design a negedge Triggered T-flipflop, we can use a total of 6 MUXES. of 2:1

- 1-mux configured to implement the XOR characteristic next-state logic ($T \oplus Q$)
- 1-mux configured as an inverter to generate the $\overline{\text{clk}}$ signal.
- 4-muxes configured in a master-slave latch architecture driven to capture data on the falling edge.
- 1-mux configured as a logic inverter to generate $\overline{Q}$.

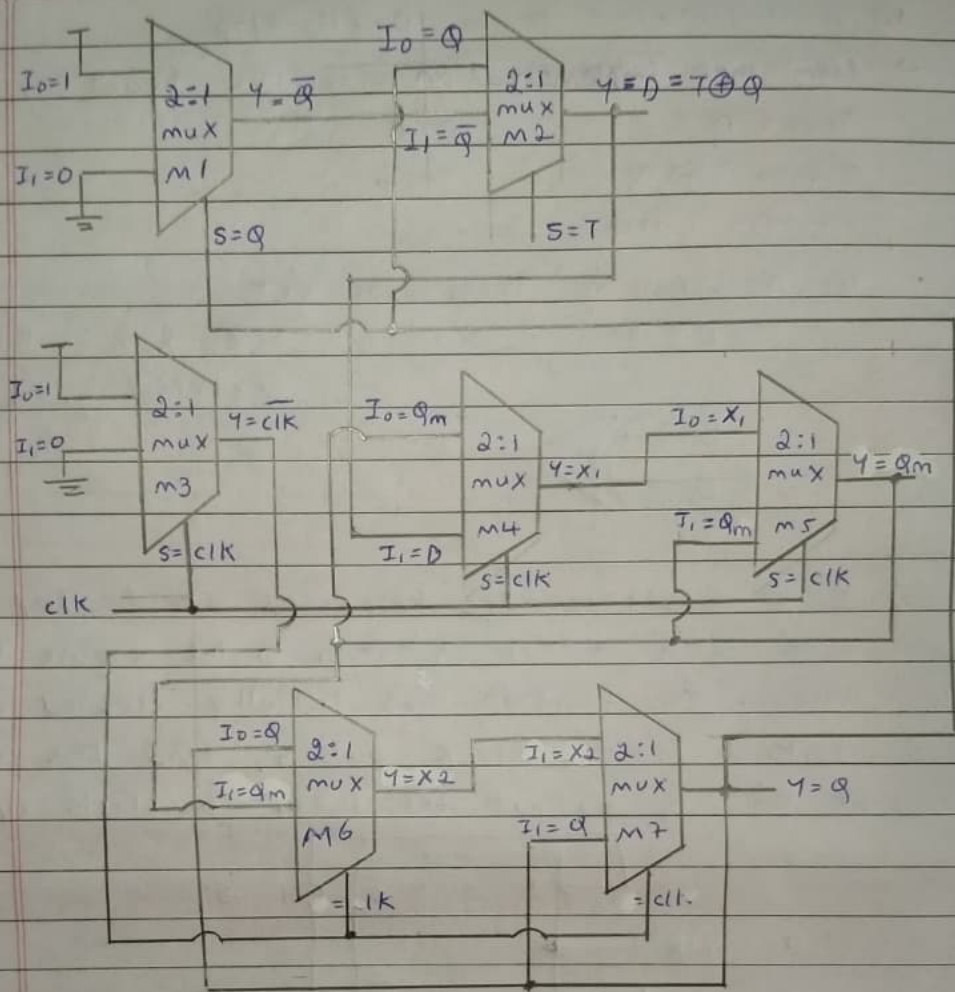
* inverter configuration: A 2:1 mux acts as a NOT gate by setting the select line to the signal to be inverted.
Input $I_0 = 1$ (VCC) & $I_1 = 0$ (GND)
output = $\overline{\text{select}} \cdot (1) + \text{select} \cdot (0) = \overline{\text{select}}$

MUX Implementation:

mux state	select line (S)	Input I_0	Input I_1	output	function
mux-1	Q	1	0	$\overline{Q}$	Q inverter
mux-2	T	Q	$\overline{Q}$	$D = T \oplus Q$	XOR logic
mux-3	clk	1	0	$\overline{\text{clk}}$	clock inverter
mux-4	clk	Q_m	D	X_1	master input
mux-5	clk	X_1	Q_m	Q_m	master feedback
mux-6	$\overline{\text{clk}}$	Q	Q_m	X_2	slave input
mux-7	$\overline{\text{clk}}$	X_2	Q	Q	slave feedback

Functional Truth Table

clock (clk)	Toggle input (T)	next state (Q_{next})	operating mode
↓ (falling edge)	0	Q	hold (o/p remains unchanged)
↓ (falling edge)	1	$\overline{Q}$	Toggle (o/p reverse state)
1 (high level)	X	Q	static hold: (master updates, slave isolated)
0 (low level)	X	Q	static hold: (master isolated, slave locked)



4) convert by adding external gates

(a) A D-flip flop to a JK flip flop

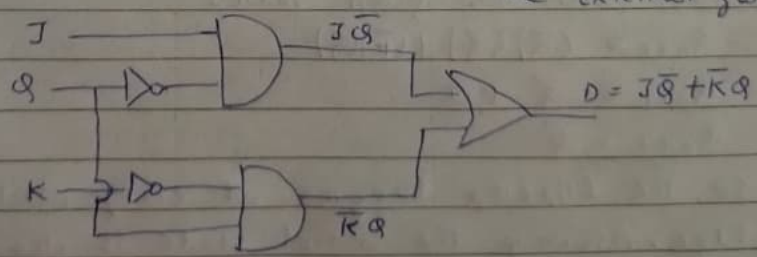
→ we know from The characteristic Equations,

$$Q_{n+1} = J \bar{Q} + \bar{K} Q \quad \text{--- JK-ff}$$

$$Q_{n+1} = D \quad \text{--- D-ff}$$

$$\therefore D = J \bar{Q} + \bar{K} Q$$

here, AND, OR, NOT are the external gates used.



(b) A T-flip flop to a D-flip flop

→ we know from the characteristic Equation,

$$Q_{n+1} = D \quad - \text{D-ff}$$

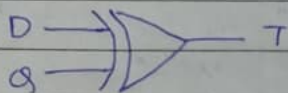
$$Q_{n+1} = T \oplus Q \quad - \text{T-ff}$$

$$\therefore D = T \oplus Q$$

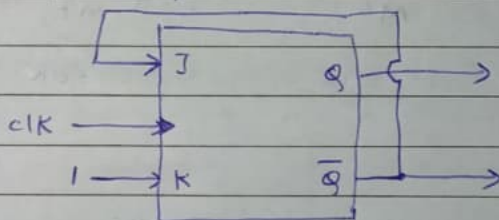
by re-arranging then, we get

$$T = D \oplus Q$$

here, XOR is the external gate used.



5) In a positive edge triggered JK flip flop, we have $J = \bar{Q}$ & $K = 1$ as shown in the figure below. Assume the flip flop was initially cleared and then clocked for 6 clock pulses, find the sequence at the Q output for those 6 clock cycles.



→ from the circuit diagram, the feedback from $\bar{Q}$ is connected to J . so,

$$J = \bar{Q}, \quad K = 1, \quad \text{initial state, } Q_0 = 0 \text{ (cleared)}$$

using JK characteristic Equation,

$$Q_{n+1} = J\bar{Q} + \bar{K}Q$$

Substitute the given values

$$Q_{n+1} = (\bar{Q})(\bar{Q}) + (\bar{1})(Q)$$

$$Q_{n+1} = \bar{Q} + 0$$

$$Q_{n+1} = \bar{Q}$$

So, the flip flop toggles at every positive clock edge. Assuming the initial state is cleared ($Q_0 = 0$)

J	K	Q_{next}
0	0	No change
0	1	0 (Reset)
1	0	1 (Set)
1	1	Toggle

state transition table

clock pulse	present Q	$\bar{Q}$	J	K	next Q
initial	0	1	1	1	-
1	0	1	1	1	1 (Toggle)
2	1	0	0	0	0 (Reset)
3	0	1	1	1	1 (Toggle)
4	1	0	0	0	0 (Reset)
5	0	1	1	1	1 (Toggle)
6	1	0	0	0	0 (Reset)

$\therefore$ The sequence of Q for 6 clock pulses :
1, 0, 1, 0, 1, 0.